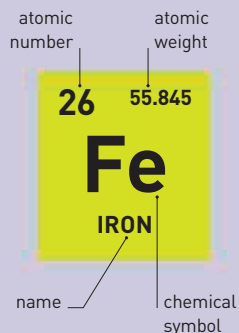




A coal train is hauled by steam locomotive *Salamanca* at the Middleton Colliery Railway, UK, in 1814.

“...RAGE FOR RAILROADS IS SO GREAT THAT MANY WILL BE LAID IN PARTS WHERE THEY WILL NOT PAY.”

George Stephenson, British civil engineer, in a letter to Joseph Sandars, December 1824



CHEMICAL SYMBOLS

The system of symbols devised by Berzelius is used by chemists even today. Each element is indicated by the initial letter or two of its Latin name—the symbol for iron, Fe, comes from its Latin name, *ferrum*. The box in the periodic table for each element shows its atomic number, its atomic weight, and the number of protons in the nucleus of each atom.

shaped like a dolphin—that lived at the time of the dinosaurs.

Also in Britain, anatomist **Charles Bell** published *Idea of a New Anatomy of the Brain*, which distinguished between the **sensory and motor nerves** of the brain.

THE FRENCH REVOLUTION'S

introduction of the **metric system** in the 1790s created chaos, as many people insisted on continuing to use the local units of measure that existed in different towns. So, in 1812, French emperor **Napoléon Bonaparte** introduced the *mesures usuelles* (standard measures), which combined basic metric units—such as the meter and kilogram—and old familiar measures. This system was finally replaced in 1840 with the full metric system.

In this year, German geologist **Friedrich Mohs** (1773–1839) devised a system to **identify minerals**. It was based on the physical properties, such as hardness, color, and shape. Mohs noticed that hard minerals could scratch softer ones. He developed a scratch test to determine the hardness of each mineral and a scale of 10 standard minerals—now known as the **Mohs scale**—on which to place each mineral. French paleontologist **Georges Cuvier** (1769–1832) published the *Discours Préliminaire* (*Preliminary Discourse*), an introduction to his essays on fossil quadrupeds (animals with



New metric system

This engraving satirizes the confusion French people had in adopting the metric system, which is why Napoleon introduced the compromise—*mesures usuelles*.

four legs), in which he argued that many more species lived on Earth in the past and that every rock bed contains fossils from a different time in Earth's past. In line with geologists who believed that the **landscapes of the world were shaped by a series of catastrophes**, Cuvier argued that the world had been overcome by past catastrophes or revolutions, which had swept away a large number of species.

Meanwhile, the **first steamboat service** in Europe opened with the paddlesteamer PS *Comet* plying on the River Clyde in Scotland. In Middleton, West Yorkshire, UK, steam locomotives were used to **haul**

trains successfully for the first time, on an adapted track initially built in the 1750s to enable horses to pull wagons full of coal from the Middleton mine.

In the 1790s, Frenchman **Nicholas Appert** had developed sealed **glass jars to preserve food**, but the glass was breakable. Then in 1810, British merchant **Peter Durand**

patented the tin can, which was made of iron coated with tin to prevent rusting. In 1812, American engraver **Thomas Kensett** (1786–1829) established the **first food preservation factory** in New York, for preserving oysters, meat, fruit, and vegetables in glass jars. In 1825, Kensett set up the first US canning factory.

James Barry (c.1792–1865)—born and raised as Margaret Ann Bulkley—chose to live as a man so she could be accepted into university. In 1812, she became the **first woman to qualify as a medical doctor**, graduating from the University of Edinburgh, Scotland. She went on to become a distinguished surgeon.

MOHS SCALE

The scale rates mineral hardness from 1 to 10 in terms of standard minerals. Geologists use scratch tests to identify a mineral—for example, one that scratches apatite but is scratched by quartz is a 6 on the scale.

1	TALC	
2	GYPSUM	
3	CALCITE	
4	FLUORITE	
5	APATITE	
6	ORTHOCLASE	
7	QUARTZ	
8	TOPAZ	
9	CORUNDUM	
10	DIAMOND	

1811 British neuroanatomist **Charles Bell** distinguishes between sensory and motor nerves

1811 French chemist **Bernard Courtois** discovers **iodine**

German geologist **Friedrich Mohs** creates his **mineral hardness scale**

February 12 **Napoleon Bonaparte** introduces the *mesures usuelles*

French palaeontologist **Georges Cuvier** argues that species have been wiped out in the past by catastrophes

August 12 First steam train runs on the Middleton Colliery line in West Yorks, UK

August 15 First steamboat service opens on the River Clyde in Scotland

Margaret Ann Bulkley—as **James Barry**—becomes the **first woman to qualify as a doctor** in Britain

American engraver **Thomas Kensett** sets up the US's first **food preservation factory**